

Tracker Tag Project

Client:

John Smith, lead asset manager from Orange Trucking Company.

Budget: The prototype costed ~\$74 to produce, but a single board device could significantly reduce the cost.

Communication:

All source code and PCB schematics will be held on a GitHub repository, alongside documentation written to help Orange's in house techs diagnose and manage devices. Technical support will be offered in a ticket-by-ticket form.

Objectives:

A low amperage, standalone tracking device that can integrate into the existing 12V system on a dry van trailer to offer reliable tracking information in case of theft or other scenario of loss.

User Interaction

Ideally, user interaction will be minimal, apart from installation and configuration. The tracker should be easily installed behind the header board, with easy

access to the 7 way cable for simple electrical hookup. Configuration should be possible remotely, once the device is assigned to an account. Configuration and location logging should be done via a simple webGUI. Trouble codes should all be available on the webGUI. In case of loss of connection, time offline should trigger an alert and log how long each device has been offline. Finally, the device should ideally never require servicing, and in case of required firmware updates, they should all be possible remotely. In case of a damaged device, schematics and troubleshooting procedures will be available on the repository, as well as a manufacturer offered repair service. These objectives will take more software optimizations, such as an integration of the API of Nextcloud Phonetrack into the GPS software to allow communication between the Pi and remote server.

Interface

The device will be remotely accessible via SSH, over the cellular network it uses for tracking.

Parameters

ECET 430 – Tracker Project Template

The device should fit in a 3"x5" rectangular space, and weigh less than 5 pounds. It will be rated IPX8 or higher for permanent exterior use, and be water, shock, and dust proof. The ratings will be with the device equipped in a case.

Functions

The device will offer accurate, up to 50m, location logging for van trailers, with the goal being to offer an effective tool to a company managing tens of thousands of units.

Integration

The device will utilize IOT LTE, 3G, and 2G networks to avoid a monthly cost per device while maintaining reliable tracking information.

Operational

The device's runtime will be restricted by how long a unit remains disconnected from an electrical system. The goal will be to have the tracker run independently for several months before requiring a charge, however the device will charge every time the unit is moved. This optimization should be done with the device's particular application chosen, and voltage source chosen to fit the application.

Life Cycle

The device will be easy and quick to produce, ideally built from easily sourced components for easy manufacturing and repair. Given that it is a tracker that will be installed in tens of thousands of units, long term support is a must. The goal will be to support these devices for a minimum of 15 years, with extended support available.

Environments

This device will be stored on the interior of a van trailer, but must be rated for constant exterior use. Temperatures will vary from 115 degrees F to -20 degrees F, as these units travel all over the North American continent.

Starting Point

This device offers similar functionality to other, vehicle specific, trackers, as well as the Apple Airtag. While the goal is to build a unique device with its own recognizable capability, the similarities must be acknowledged. The device will utilize open source tracking software and GSM/GPRS drivers.

Key Concerns

The device must be able to continue tracking without power for a certain amount of time, otherwise the usefulness of the tracker will be limited. Additionally, the device must be installed in such a way that it is not easily tampered with, and notifies an admin if it is being tampered with.

Conclusions

For a successful deployment, the device will need more development, specifically in the areas of having the GPS location reliably be obtained, and transmitted to the cloud. Furthermore, to have the device actually function, testing should be done to determine voltage and amperage ratings, and recommended pairings with voltage sources in several different scenarios. In terms of hardware, integrating all the components onto a single board could significantly decrease both footprint and price (depending on producer), which is important in several weather and space sensitive deployments, leading to the final consideration, a case. The case should be accessible as a part of the project files, and be able to endure impact and adverse weather conditions. As it stands now, to implement this device would require significant work which would dwarf the price of reverting to another commercial solution, and a core goal of this project is to compete with those same commercial solutions. While the tracker needs more development, it can do so publicly as the designs and

software are all open source. Once the product reaches maturity, I believe it will be able to compete with those solutions, and be able to do it better than if it was closed source and proprietary.